

Project Write-Up: Implementation of PDF Q&A Chatbot using RAG

July 12, 2026

1 Chosen Scope and Why

I have chosen **Academic Assistant** — **course registration, grading policy, academic calendar, exam rules** because it was a lot easier to find its documentation and verify the thing. No other reason as such.

2 List of Data Sources

- Punishments101521July
- Procedures201521July
- Student-medical-rules
- UG_Rule_Book
- Anti_Discriminatory_Guidelines

3 Chunking Strategy and Justification

I have chosen a **fixed-size chunking mechanism (200 characters with a 40-character overlap)** because passing an entire PDF document to the AI at once just causes memory overflows and hits max token limits. We are using the sentence-transformers/all-MiniLM-L6-v2 model, which has a strict 256-token limit, so 200 characters keeps us safe from abrupt cut-offs. Also, smaller chunks keep the vectors concentrated so the FAISS database can actually find the right paragraph. The 40-character overlap is just a safety net so we don't accidentally chop a compound sentence or important keyword in half. No other major reasons for this setup.

4 Known Limitations and Future Upgradations

While the bot works for basic text, there are a few main issues right now:

- **Complex Data:** PyPDF2 completely fails on scanned PDFs, images, and tables.

- **No Memory:** The system treats every question as brand new, making follow-ups difficult.
- **Basic Retrieval:** If the first FAISS search misses the right chunk, the LLM just refuses to answer.

If I had more time to upgrade it, I would definitely plug in OCR to read scanned documents and tables. I'd also swap the basic chunking for Semantic Chunking to respect actual paragraph boundaries, and add a database for chat histories to make it feel like a real conversation.